

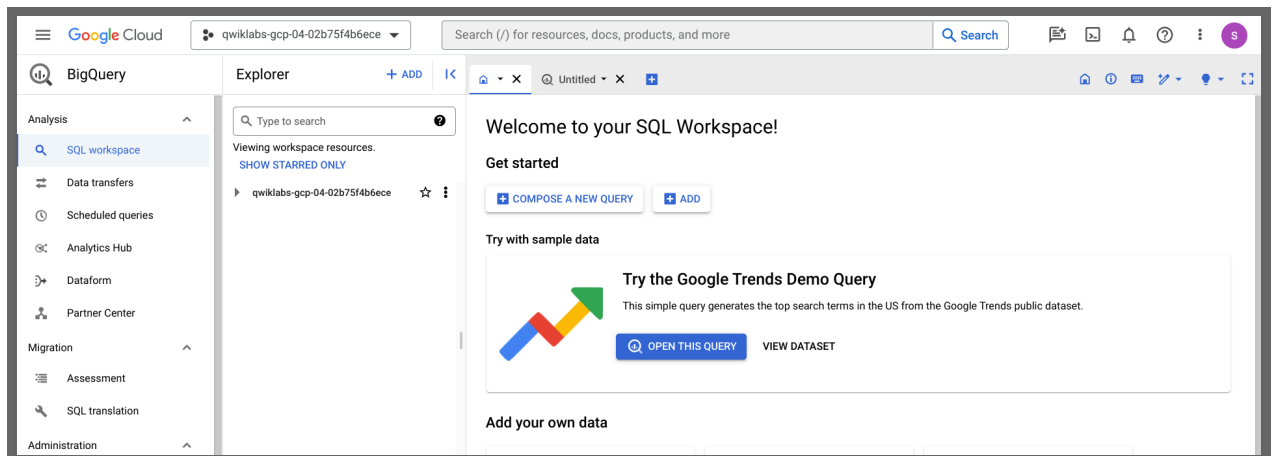
Create a partitioned table

Previously, you learned that there are three primary types of table partitioning: integer-range, time-unit, and ingestion-time. Integer-range partitioning divides tables into partitions based on ranges of values in a specific integer column. Time-unit partitioning divides tables into partitions based on a date or timestamp column. Ingestion-time partitioning divides tables into partitions based on when the data was ingested into the table. In this reading, you'll learn how to create integer-range partitions and time-unit partitions.

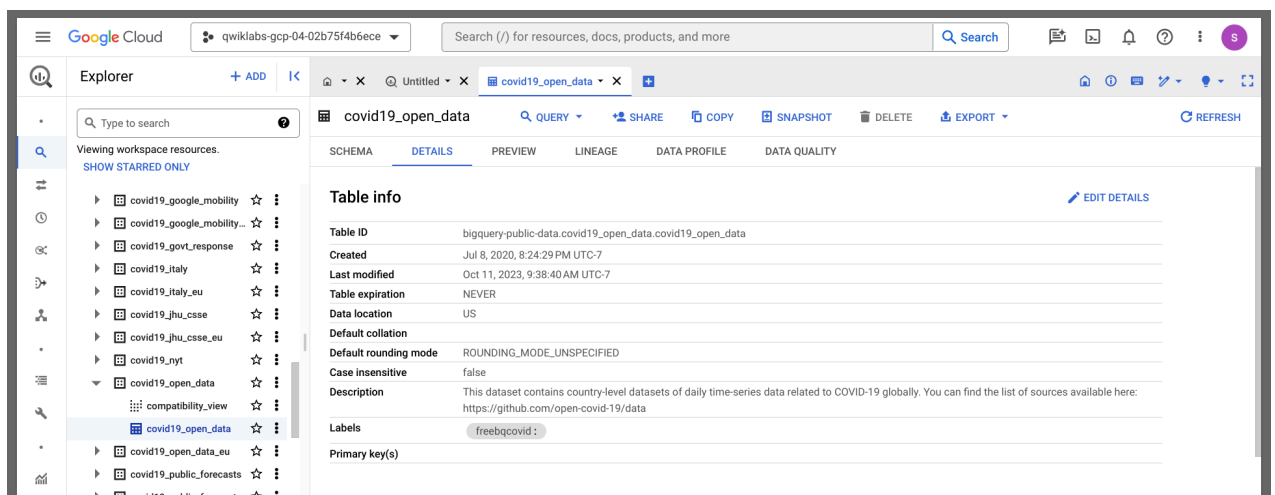
Create a table

The initial steps for creating a partitioned table are identical for all types of partitions. From the Google Cloud console, navigate to the BigQuery page. Then, follow these steps:

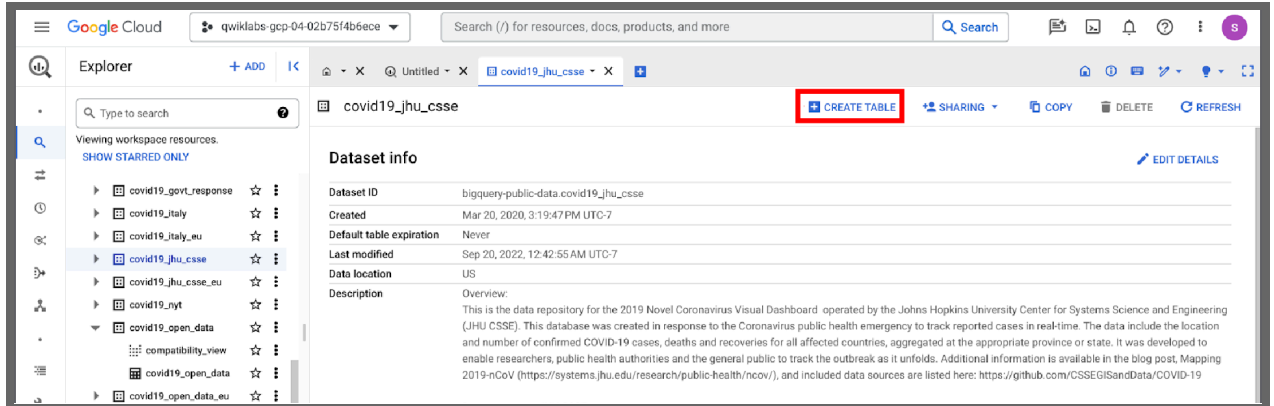
1. Navigate to the Explorer panel.



2. Select and open your dataset, and navigate to the details page.



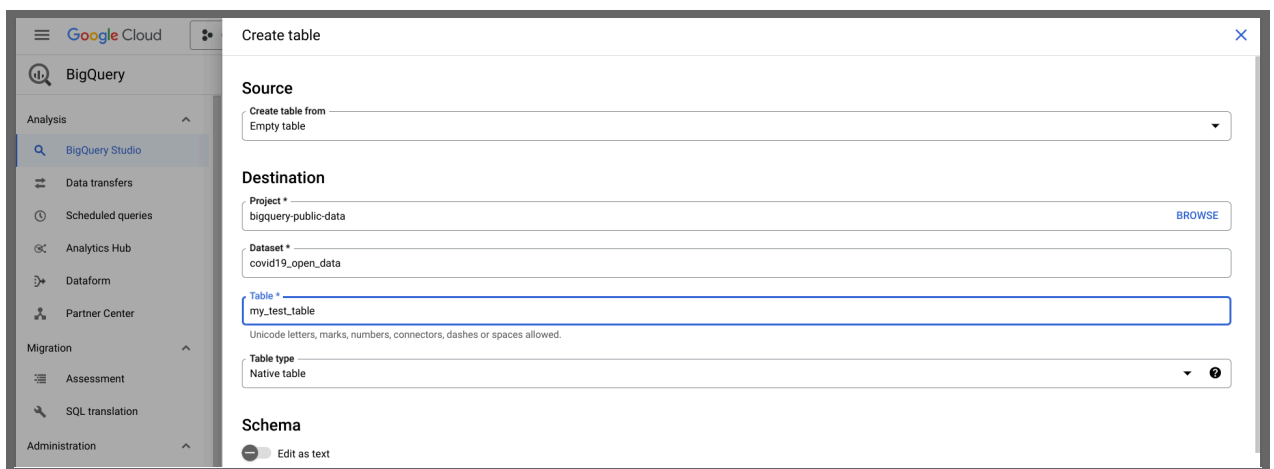
3. Click CREATE TABLE.



4. In the Source section, select Empty table.



5. Fill in the Dataset name and Table name.



Once you've created your table, the next steps will determine if it's an integer-range partition or a time-unit partition.

Create an integer-range table

Follow these steps to create an integer-range table:

1. Enter your Schema and ensure that you have an **INTEGER** column.

Create table

Table type
Native table

Schema

Edit as text

Field name *	Type *	Mode	Description
1 INTEGER	INTEGER	NULLABLE	
2 Name	STRING	NULLABLE	Max len... Description
3 ID	STRING	NULLABLE	Max len... Description

4 +

Partition and cluster settings

Partitioning
Partition by ingestion time

Partitioning filter ⓘ
☐ Require WHERE clause to query data

Partitioning type ⓘ
☐ By day

CREATE TABLE CANCEL

2. Navigate to Partition and cluster settings.

Create table

bigquery-public-data **BROWSE**

Dataset *
covid19_jhu_csse

Table *
Unicode letters, marks, numbers, connectors, dashes or spaces allowed.

Table type
Native table

Schema

Edit as text

+

Partition and cluster settings

Partitioning
No partitioning

Clustering order ⓘ
Clustering order determines the sort order of the data. Clustering can be used on both partitioned and non-partitioned tables.

Advanced options ▼

CREATE TABLE CANCEL

3. In the Partition and cluster settings drop-down, select Partition by field: INTEGER.

Google Cloud Create table

Field name * ID Type * STRING Mode * NULLABLE Max len... Description

Partition and cluster settings

Partitioning

Options

No partitioning

Partition by ingestion time

Partition by field

INTEGER

By month

By year

4. Enter the Start, End, and Interval for each partition.

Google Cloud Create table

Partition and cluster settings

Partitioning

INTEGER

Partitioning filter

Require WHERE clause to query data

Start *

Range start is required

End *

Range end is required

Interval *

Range interval is required

Clustering order

Clustering order determines the sort order of the data. Clustering can be used on both partitioned and non-partitioned tables.

Advanced options

Encryption

Google-managed encryption key

No configuration required

Customer-managed encryption key (CMEK)

CREATE TABLE CANCEL

Note: Start the start of the partition range, End is the end of the last partition range, and Interval is the width of each partition range.

5. Click CREATE TABLE

Create table

Partitioning
No partitioning

Clustering order
Clustering order determines the sort order of the data. Clustering can be used on both partitioned and non-partitioned tables.

Advanced options

Encryption

☒ Google-managed encryption key
No configuration required

☐ Customer-managed encryption key (CMEK)
Manage via [Google Cloud Key Management Service](#)

Default Collation

☒ Enable default collation

Default Collation

Default Rounding Mode
If not set, the table default rounding mode will be inherited from the dataset and will apply to all DECIMAL (numeric and bignumeric) columns created. This setting can be created or altered later and if altered, will only apply to columns added after the setting is changed.

Default Rounding Mode

CREATE TABLE **CANCEL**

Create an ingestion-time partitioned table

Once you've followed the first five steps to create a table in BigQuery, create a time-unit partition:

1. Navigate to Partition and cluster settings. In the Partitioning section, click Partition by ingestion time.

Create table

Table type
Native table

Schema

☒ Edit as text

Partition and cluster settings

Partitioning

Options

☒ No partitioning

☐ Partition by ingestion time

Partitioning by

☐ By day

☐ By hour

☐ By month

☐ By year

Clustering order
Clustering order determines the sort order of the data. Clustering can be used on both partitioned and non-partitioned tables.

Advanced options

CREATE TABLE **CANCEL**

2. Click CREATE TABLE.

Create a time-unit table

If you need to create a time-unit table, create an empty table first, then:

1. Enter your Schema and ensure that you have a **DATE, TIMESTAMP or DATETIME** column.

2. Navigate to Partition and cluster settings.

Create table

bigquery-public-data BUKWSZ

Dataset *
covid19_jhu_csse

Table *
Unicode letters, marks, numbers, connectors, dashes or spaces allowed.

Table type
Native table

Schema
Edit as text

Partition and cluster settings
Partitioning
No partitioning
Clustering order
Clustering order determines the sort order of the data. Clustering can be used on both partitioned and non-partitioned tables.

Advanced options

CREATE TABLE **CANCEL**

3. In the Partitioning drop-down, select Partition by field: DATE

Create table

Field name *
ID

Type *
STRING

Mode
NULLABLE

Max len...
Description

Partition and cluster settings
Partitioning
Options
No partitioning
Partition by ingestion time
Partition by field
INTEGER
By month
By year
Clustering order
Clustering order determines the sort order of the data. Clustering can be used on both partitioned and non-partitioned tables.

Advanced options
Encryption
Google-managed encryption key

CREATE TABLE **CANCEL**

4. Select the Partitioning type: By day.

Create table

Schema

☐ Edit as text

	Field name *	Type *	Mode	Description
1	INTEGER	INTEGER	NULLABLE	
2	DATE	DATE	NULLABLE	
3				

Partition and cluster settings

Partitioning: DATE

Partitioning filter [?]

☐ Require WHERE clause to query data

Partitioning type [?]

☒ By day

Clustering order [?]

Clustering order determines the sort order of the data. Clustering can be used on both partitioned and non-partitioned tables.

Advanced options [?]

CREATE TABLE **CANCEL**

5. Click CREATE TABLE.

Create table

Partition and cluster settings

Partitioning: No partitioning

Clustering order [?]

Clustering order determines the sort order of the data. Clustering can be used on both partitioned and non-partitioned tables.

Advanced options [?]

Encryption [?]

☒ Google-managed encryption key
No configuration required

☐ Customer-managed encryption key (CMEK)
Manage via [Google Cloud Key Management Service](#)

Default Collation

☒ Enable default collation [?]

Default Collation

Default Rounding Mode

If not set, the table default rounding mode will be inherited from the dataset and will apply to all DECIMAL (numeric and bignumeric) columns created. This setting can be created or altered later and if altered, will only apply to columns added after the setting is changed.

Default Rounding Mode

CREATE TABLE **CANCEL**

Determine whether a table is partitioned

If you have an existing table, and want to determine whether it's a partitioned table, you can look at the table details in the User Interface (UI) which will include the partition specification if there is one. The table details show if it uses ingestion time, time unit, or integer range partitioning, and the column it's partitioned on.

Key takeaways

Integer-range and time-unit partitioned tables can divide your data tables by specific values present in the data, while ingestion-time partitioned tables can divide your data by the time the data is ingested. All types of partitions can be managed from the Google Cloud console using BigQuery by navigating to your dataset, and selecting the appropriate partition type when creating the table.

Resources for more information

Visit the link to review Google Cloud's documentation for creating partitioned tables in BigQuery:

- Google Cloud provides step-by-step instructions for creating partitions in the console, via SQL, and with Java or Python:

<https://cloud.google.com/bigquery/docs/creating-partitioned-tables>